

DARPA DISCORD

Disruption through Intelligent Strategies, Counter Options, and Resilient Defenses

OUSW(R&E) Ender's Foundry



Eric Davis, DARPA STO

2026/05/07





Schedule of Proposer's Day



Start	End	Event
09:30	9:40	Security and Logistics
9:40	10:20	DISCORD Overview
10:20	10:35	Stakeholder Brief
10:35	10:50	Stakeholder Brief
10:50	11:05	Proposal Submission Guidance
11:05	12:30	Submit questions / LUNCH
12:30	12:45	DARPA Connect
12:45	15:15	Lightning Talks
15:15	15:30	BREAK
15:30	16:00	Answer Questions / Closing Remarks
16:00		Optional No Host Happy Hour @ El Rey (4201 Wilson Blvd, Arlington, VA 22203)



Context | Department of War AI Acceleration Strategy



OUSW(R&E) and CDAO are partnering for a number of pace-setting projects intended to be outcome-oriented vehicles for rapid development of enabling foundational AI capabilities.

DISCORD (referred to as Ender's Foundry by OUSW(R&E) right) is intended to provide capabilities to stay ahead of AI-enabled adversaries in warfighting.



SECRETARY OF WAR
1000 DEFENSE PENTAGON
WASHINGTON, DC 20301-1000

JAN - 9 2026

MEMORANDUM FOR SENIOR PENTAGON LEADERSHIP
COMMANDERS OF THE COMBATANT COMMANDS
DEFENSE AGENCY AND DOW FIELD ACTIVITY DIRECTORS

SUBJECT: Artificial Intelligence Strategy for the Department of War

Accelerating America's Military AI Dominance

President Trump makes clear in Executive Order 14179, "It is the policy of the United States to sustain and enhance America's global Artificial Intelligence (AI) dominance in order to promote

In the national security will re-define the race — fueled by America's private adversaries in intelligence make our Warfighting Action Plan, I direct by becoming an "

The Department

- Unleash wide,
- Aggregate intelligence warfighting
- Focus AI combat operations
- Execute acceleration

will also ensure the Department's success in rapidly expanding the buildup of the foundational AI enablers (infrastructure, data, models, policies, and talent) needed to accelerate AI integration across the entire Department.

• Warfighting:

1. **Swarm Forge:** Competitive mechanism to iteratively discover, test, and scale novel ways of fighting with and against AI-enabled capabilities — combining America's elite Warfighting units with elite technology innovators.
2. **Agent Network:** Unleashing AI agent development and experimentation for AI-enabled battle management and decision support, from campaign planning to kill chain execution.
3. **Ender's Foundry:** Accelerating AI-enabled simulation capabilities — and sim-dev and sim-ops feedback loops — to ensure we stay ahead of AI-enabled adversaries.

Figure Source: <https://media.defense.gov/2026/Jan/12/2003855671/-1/-1/0/ARTIFICIAL-INTELLIGENCE-STRATEGY-FOR-THE-DEPARTMENT-OF-WAR.PDF>

AI: Artificial Intelligence

OUSW(R&E): Office of the Under Secretary of War for Research and Engineering

CDAO: Chief Digital and Artificial Intelligence Office

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Translating OUSW(R&E) Intent to DARPA Impact



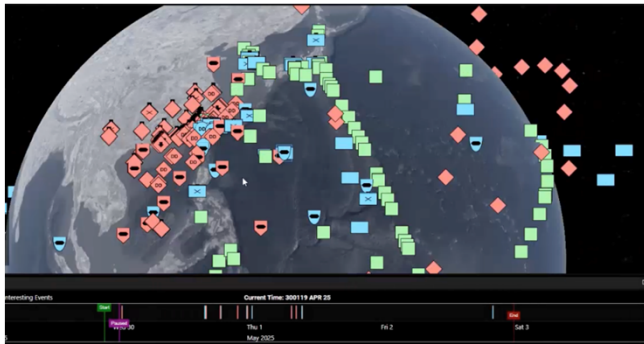
OUSW(R&E) Ender's Foundry Goals

- Ensure US stays ahead of AI-enabled adversaries
- Build on success of SCEPTER: accelerate M&S environments to 100,000x real-time to support real-time execution
- Generate a tactics "factory" capable of executing 10,000+ tactical experiments daily
- Develop a counter-UxS proving ground
- Exploit vulnerabilities in adversarial COAs and architecture at machine speed

DARPA DISCORD Goals

- Collapse deliberate planning and execution to a new AI-native approach suited to the tempo of adaptation fights
- Identify non-linear "team-breaking" vulnerabilities in adversarial mission architectures
- Furnish commanders with a portfolio of strategies that offer options to a wide range of possible futures

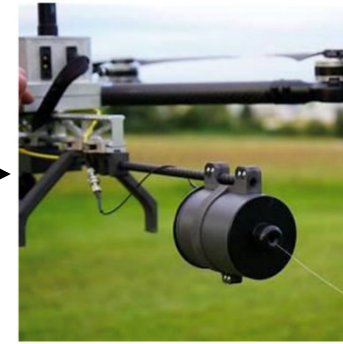
Phase 1: Simulated Proving Ground



OUSW(R&E): Office of the Under Secretary of War for Research and Engineering
AI: Artificial Intelligence

SCEPTER: Strategic Chaos Engine for Planning, Tactics, Experimentation and Resiliency
M&S: Modeling and Simulation

Phase 2: LVC Proving Ground



UxS: Unmanned Systems
COAs: Courses of Action
LVC: Live, Virtual, Constructive

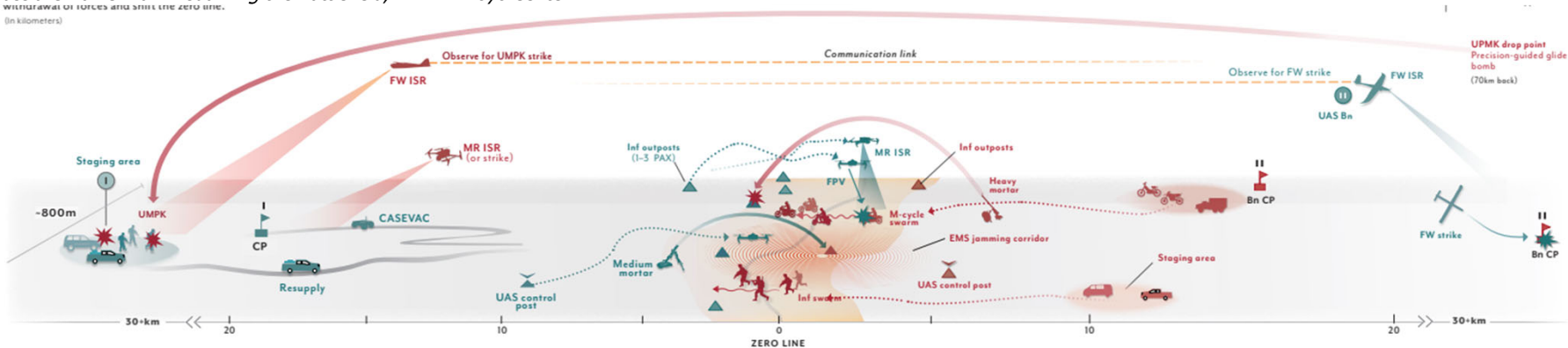
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Current Planning Paradigm is Outpaced and Outmatched



Russia-Ukraine War: Visualizing the Battlefield, RAND Arroyo Center
WESTSTAR ARROYO FOR APODOLIS ARROYO ARROYO ARROYO ARROYO
(in kilometers)



The Russo-Ukrainian War provides daily evidence that the character of conflict has changed. Adversaries are demonstrating an ability to adapt tactics and technology in under two weeks. The deliberate planning process fails to match this tempo. This creates three critical limitations.

- | | |
|------------------------|---|
| Tactical Brittleness | Doctrine fails to adapt with the battlefield. Sophisticated air defenses are consistently overwhelmed by saturation attacks from low-cost drones. A single interceptor can cost 10-100x more than its target. Static, kinetic, attrition-based planning is unsustainable. |
| Shallow Insight | In Ukraine, both sides have shown that success hinges on identifying and exploiting weaknesses in an adversary's communications, sensing, or C2. However, these discoveries are ad-hoc and opportunistic. |
| The Illusion of Choice | Adversaries in Ukraine adapt methods of attack in hours. The deliberate process delivers a single, "optimized" COA to commanders with no time to formally explore or validate a portfolio of options in response to real time data on a dynamic enemy. |

C2: Command and control
COA: Course of Action

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DISCORD program goals



DISCORD aims to create an AI-Native process collapsing planning and execution, to out-pace and out-think adaptive adversaries in real-time.

TECHNICAL AREAS

CHALLENGES

APPROACH

TA 1: Asymmetric Advantage Engine

Current surrogate model acceleration is insufficient for real-time replanning. Current models are slow to retrain and assume multiple training examples to adapt.

Explore new methods for accelerated, compositional function lifting. Incorporate non-linear time series analysis methods to learn from fewer examples and single-shot.

TA 2: Human-Machine Teaming

AI-Native processes move at machine speed but are bottlenecked by the human components of decision and execution due to the limits of explainable AI.

Alternatives to generating a universally explainable AI or interface. Learn profiles of individuals through limited engagements enabling adaptation to specific humans.

TA 3: Live Adaptation Experimentation

Simulation fails to address warfighter skepticism and sim-to-real gap, including integration uncertainties.

Integrate DISCORD into remote procedure and object frameworks and conduct LVC testing with warfighters.

C2: Command and Control
AI: Artificial Intelligence
TA: Technical Area
LVC: Live, Virtual, and Constructive

SCEPTER: Strategic Chaos Engine for Planning, Tactics,
Experimentation and Resiliency
APIs: Application Programming Interfaces

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Takens' Delay Embedding Theorem (1981)

Let a d -dimensional dynamical system evolve on a smooth attractor, and let $y(t)$ be a generic smooth scalar observation of the system.

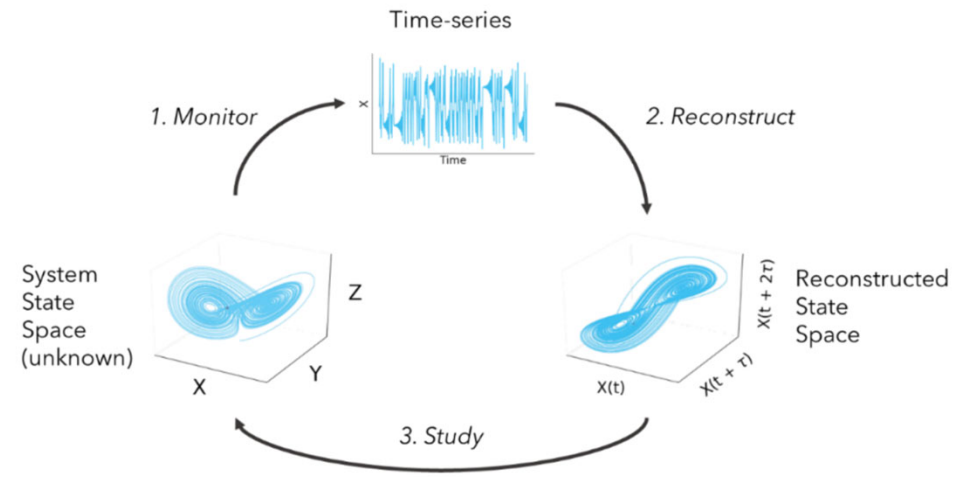
Then the delay embedding map

$$\mathbf{Y}(t) = [y(t), y(t - \tau) \dots y(t - (k - 1)\tau)]$$

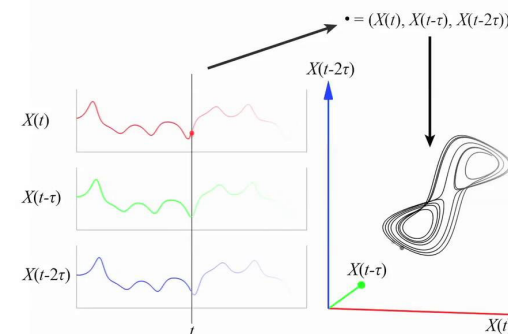
is an embedding (i.e., a one-to-one, smooth reconstruction of the attractor) for $k \geq 2d + 1$.

Thus, the geometry of the original attractor (e.g., the Lorenz attractor) can be reconstructed from time-delayed observations of a single variable.

Takens' theorem shows that a single observed time series implicitly contains the full state information via time delays. Unlike Reinforcement Learning or Inverse Reinforcement Learning, you don't need many trajectories or rich observations to reconstruct the underlying dynamics.



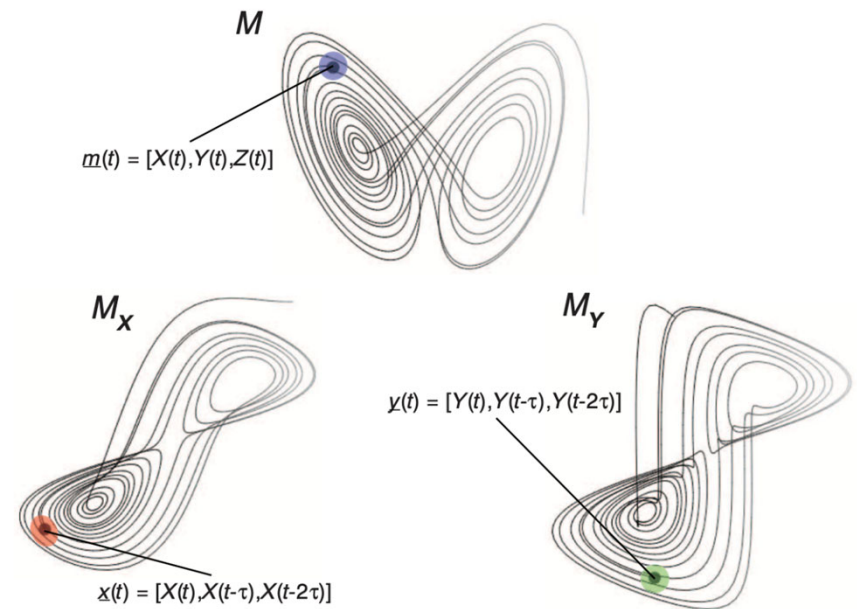
Delforge, 2020



Sugihara et al., 2020

Using Takens' embedding theorem, we reconstruct separate delay-embedded manifolds from each observed time series (e.g., M_X from $X(t)$, M_Y from $Y(t)$). (If the variables are part of the same dynamical system, these manifolds are different projections of the same underlying attractor. As a result, their geometries are linked: neighborhoods on one manifold correspond to neighborhoods on the other).

Think of each delay-embedded manifold as a geometric "fingerprint" of a variable's history. To test causality, we pick a point on M_Y , find nearby points (similar pasts of Y), and look at the corresponding times in X . If those X values are also similar, it means Y 's history reliably tells us what X was doing. When this reconstruction gets better with more data (and works in one direction more than the other) it indicates a causal influence.



Sugihara et al., 2020



TA 1 – Asymmetric Advantage Engine



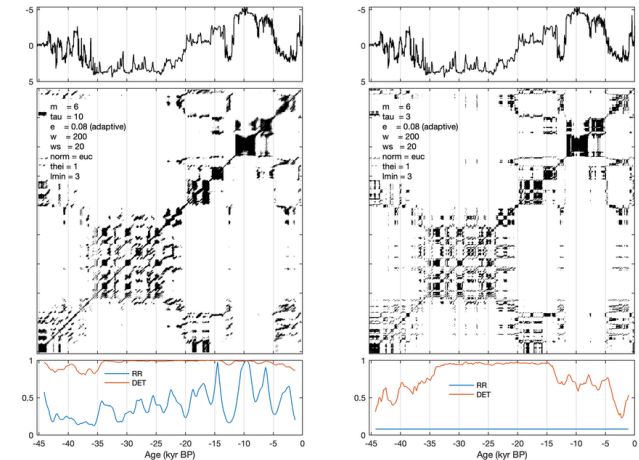
Challenge: The SCEPTER program succeeded in accelerating physics-based modeling. While suitable for deliberate planning, it is insufficient for real-time replanning in adaptive environments.

Insight: DISCORD will learn from the geometry of joint system behavior. Using nonlinear time series techniques, based on *Takens' Embedding Theorem* (such as RQA), TA 1 will reconstruct an adversary's dynamic state from short windows of simulated and live sensor data, revealing hidden features while reducing the training time and data requirements by up to three orders of magnitude.

DISCORD solutions will reason in advance to generate a portfolio of strategies as a starting point and will then adapt estimated models with real data to refine COAs in the portfolio accounting for adversarial surprise.

Approach and Metrics: TA 1 solutions will be given commander's intent from TA 2 to optimize against, using surrogate models as approximate policy oracles.

Successful TA 1 solutions will not only be faster than real time but will use that speed up to efficiently search for tactics which use recovered **architectural insight** of vulnerabilities to generate COAs which maximize commander's intent providing **advantage** over baseline COAs. Successful



Trauth et al. 2019

RQA is an advanced nonlinear data analysis technique that is often described as "data-efficient". RQA reveals new features like determinism, recurrence rate, and laminarity between interacting components. RQA has shown more than a 100x improvement when used to enhance inverse reinforcement learning.

Metrics	Phase 1	Phase 2
Advantage Score – JP 5-0 weighted numerical comparison	>10	>100
Architectural Insight – Percentage of pre-identified vulnerabilities detected	>75%	>98%

TA: Technical Area

SCEPTER: Strategic Chaos Engine for Planning, Tactics, Experimentation and Resiliency

RQA: Recurrence Quantification Analysis

COAs: Courses of Action

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TA 2 – Human-Machine Teaming



Challenge: AI-Native processes move at machine speed enabling rapid discovery of dominating tactics in controlled experiments. Command post exercises with the SCEPTER tool have revealed the growing human bottleneck. Operators must iterate to translate commander's intent into the tool to produce COAs and then work to understand the outputs.

Insight: DISCORD will create an AI that understands its role within a human-machine team. Instead of just presenting data, the system will actively modeling the team's shared goals, skills, and dependencies; including each individual's expertise and knowledge. Interdependence analysis and transactive memory systems have shown the ability to improve machine common ground understanding, and improve teaming outcomes, including tempo.

The result is a true cognitive partner that makes the human commander more powerful, not a complex tool that overwhelms them; and learns individual warfighters in **no more than 25-50 interactions** as achieved during initial deliberate exploration during evaluation.

Approach and Metrics: TA 2 solutions will work with minimal data from warfighters using the system and build profiles rapidly.

Successful TA 2 solutions will have a measurable impact on the **speed of command**, allowing warfighters to come to decisions with DISCORD in 60s or less. They will do so by achieving >95% **intent translation accuracy** of natural language into weighted JP 5-0 criteria for TA 1.

Example Numerical Comparison

		Course of Action					
		COA 1		COA 2		COA 3	
Criteria	Weight	Rating	Product	Rating	Product	Rating	Product
Exploits maneuver	2	3	6	2	4	1	2
Attacks COGs	3	2	6	3	9	1	3
Integrates maneuver and interdiction	2	2	4	3	6	1	2
Exploits deception	2	1	2	2	4	3	6
Provides flexibility	2	1	2	3	6	2	4
CSS (best use of transportation)	1	3	3	2	2	1	1
Total		12		15		9	
Weighted total			23		31		18

NOTE: The higher the number, the better.

An example of the COA comparison process from JP 5-0. Commanders must first establish criteria and weightings for those criteria to numerically compare COAs. Each commander will define different criteria and weighting, meaning there is never one best COA from an objective basis.

Metrics	Phase 1	Phase 2
Speed of Command – Time required for a commander to orient to a tactical problem, comprehend options, and make a confident decision	< 5m	< 60s
Intent Translation Accuracy – Ability to correctly interpret commanders intent, automating weighting of JP 5-0 criteria	>80%	>95%

TA: Technical Area
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COAs: Courses of Action

JP 5-0: Joint Publication 5-0, *Joint Planning*, dated 16 June 2017
COGs: Centers of Gravity
CSS: Combat Service Support

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TA 3 – Live Adaptation Experimentation



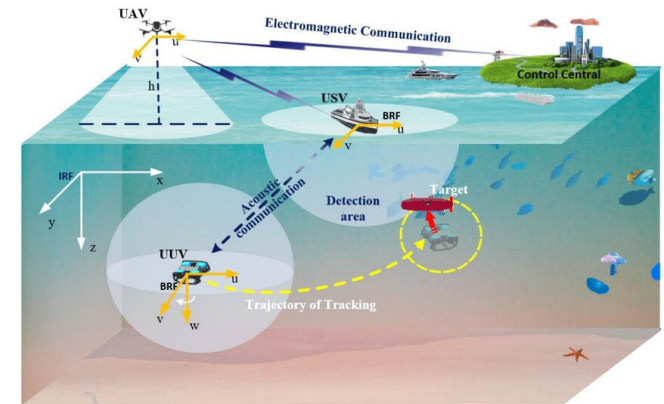
Challenge: DISCORD’s envisioned AI-native collapse of planning and C2 cannot be proven in simulation alone. Simulated exercises fail to address warfighter skepticism and the sim-to-real gap, including integration uncertainties.

Insight: DISCORD will integrate TA 1 and TA 2 solutions into real world capabilities in a LVC proving ground. Live sensing and drones will be combined with simulated electronic warfare and kinetic effectors such as close in weapons systems. TA 3 focuses on moving DISCORD from a theoretical engine to one ready for validation by warfighters.

TA 3 focuses on the engineering requirements for integration of the software and hardware components with DISCORD’s partners as part of regularly scheduled demonstrations and exercises.

Approach and Metrics: TA 3 solutions must manage the integrated system’s real time planning requirements efficiently in simulated and LVC experiments.

Successful TA 3 solutions will ensure that **time to effective response** of the integrated prototype adds no more than additional 60s to TA 2 metrics; while iterating with TA 1 components over live data to ensure **plan resilience under fire** of 80% of the COAs for a commander’s intended advantage.



DISCORD will be tested in multi-domain UxS proving grounds with live, virtual, and constructive components. Existing testing facilities already implement live sensing capabilities that are fused and made available to Maven Smart Systems as standardized object models and feature large-scale UxS capabilities including aerial, surface, sub-surface, and ground capabilities that respond to standard remote procedure call systems like gRPC.

Metrics	Phase 1	Phase 2
Time to Effective Response – End to end battlefield agility. Total time to sense plan-invalidating surprise through execution of new counter-plan	< 10m	< 2m
Plan Resilience Under Fire – Percentage of the original portfolio still viable after a sustained period of live engagement	> 50%	> 80%

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COAs: Courses of Action
UxS: Unmanned System
gRPC: Google Remote Procedure Call

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DISCORD Metrics



The DISCORD metrics are designed to provide quantitative, warfighter-oriented evidence that the program is systematically retiring the primary risks of the program.

Scheduled down selects will occur for failure to meet conditions of integration (Months 3, 6, and 18) or for failure to meet the specified metrics per TA (Months 12, 24).

Regular command post exercises will be executed as available with INDOPACOM and USSOCOM by the government team with performer technologies.

Metrics		Phase 1	Phase 2
TA 1	Advantage Score – JP 5-0 weighted numerical comparison	>10	>100
	Architectural Insight – Percentage of pre-identified vulnerabilities detected	>75%	>98%
TA 2	Speed of Command – Time required for a commander to orient to a tactical problem, comprehend options, and make a confident decision	< 5m	< 60s
	Intent Translation Accuracy – Ability to correctly interpret commanders intent, automating weighting of JP 5-0 criteria	>80%	>95%
TA 3	Time to Effective Response – End to end battlefield agility. Total time to sense plan-invalidating surprise through execution of new counter-plan	< 10m	< 2m
	Plan Resilience Under Fire – Percentage of the original portfolio still viable after a sustained period of live engagement	> 50%	> 80%

TA: Technical Area
LVC: Live, Virtual, and Constructive
INDOPACOM: US Indo-Pacific Command
USSOCOM: US Special Operations Command

JP 5-0: Joint Publication 5-0, Joint Planning,
dated 16 June 2017



DISCORD Government Team



The DISCORD government team is organized to steer, ground, and validate the program from inception to transition. The team's structure combines operational leadership with deep technical expertise to guide performers and independently verify results.

The government advisory and experimentation team will define the critical c-UxS problem sets, platforms, capabilities, and adversaries for DISCORD's tests.

IDA will translate these operational requirements into specific, intellectually challenging scenarios engineered to probe the limits of performers' systems.

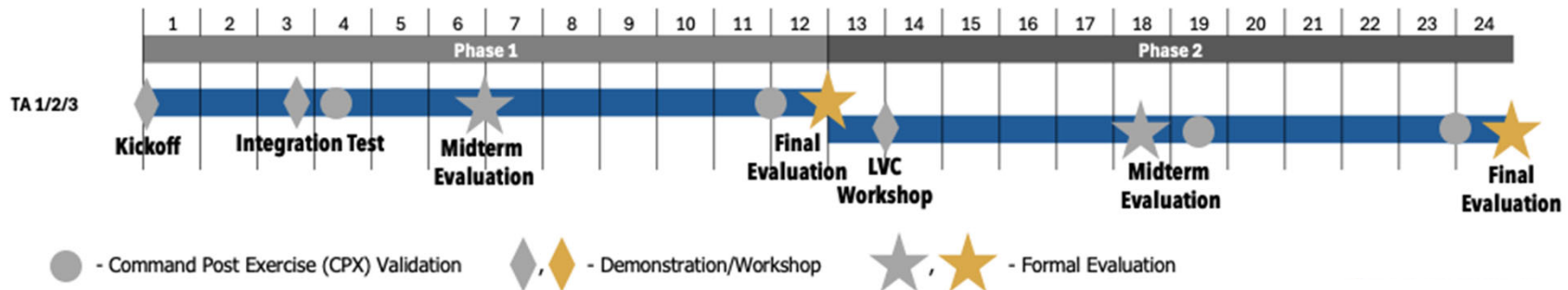
MITRE will implement these scenarios in their SIMEX environment already used in UxS exercises by key combatant commands.

The DISCORD government team will build and implement these scenarios within their advanced LVC testbeds.

c-UxS: Counter Unmanned Systems
IDA: Institute for Defense Analysis
SIMEX: Simulation Experiment
UxS: Unmanned System
LVC: Live, Virtual, and Constructive



DISCORD Program Schedule



- (Month 3) Integration Test:** At program kickoff, the Government Team will provide all performers with an integrated simulation environment, a baseline c-UxS scenario, and initial APIs for connecting TA 1 and TA 2 systems. By Month 3, all TA 1 and TA 2 performers must demonstrate a "steel thread" proof of concept, showing an end-to-end workflow that successfully exercises these APIs.
- (Month 6) Midterm Evaluation:** TA 1 and TA 2 performers will execute a dry run of the full evaluation process, demonstrating a Minimum Viable Product of their integrated capabilities within a live simulation that includes synthetic data injections of novel confounds. TA 3 performers will demonstrate a candidate system capable of receiving and acting upon TA 2 command structures in a live (hardware) demonstration.
- (Month 12) Phase 1 Formal Evaluation:** TA 1 and TA 2 performers will conduct a formal evaluation of their integrated systems, demonstrating quantitative progress against the Phase 1 program metrics. TA 3 performers will execute a live flight demonstration, showing their ability to adapt in real-time to simulated commands and achieve the required Phase 1 metrics. This event serves as the basis for down-selection to Phase 2.
- (Month 13) Phase 2 Kick-Off:** Successful performers will participate in a joint kick-off and workshop at a government test facility, interacting with joint UAS, USV, and UUV platforms to build a concrete technical roadmap for the final phase of the program.
- (Month 18) End-to-End Integration Demonstration:** Performers will demonstrate a fully integrated TA 1, TA 2, and TA 3 system operating in a joint UxS scenario at a government test facility, proving the end-to-end functionality of their solution.
- (Month 24) Final Graduation Exercise:** The program will culminate in a final, scored "graduation" exercise in a full LVC environment orchestrated by an organization on the government team. This final exam will include novel adversary tactics developed by IDA, live human operators from combatant commands, and formal scoring of the final Phase 2 metrics by MITRE to determine ultimate program success.



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